Models

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29th October 2022

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First-order structures

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Definition

A *structure* for a first-order signature Σ is a tuple $\langle D, (I_s)_{s \in R_\Sigma}, (I_t)_{t \in F_\Sigma} \rangle$, where:

- D (called the "domain" of the structure) is not empty
- $I_s \subseteq D^n$ when $s \in R_\Sigma$ and $a_\Sigma s = n$
- $I_t \in D^{D^n}$ when $t \in F_\Sigma$ and $a_\Sigma s = n$.

Definition

When S is a structure for Σ with domain D, an *assignment function for S* is a function from Var to D.

Truth in a structure on an assignment

We define a notion of a formula P being **true in** a given structure S on a given assignment g.

- $S, g \Vdash P$ means 'P is true in S on g'.
- We also write this as '$g \in [\![P]\!]_S$'.
- Or as '$[\![P]\!]_S^g = 1$'.

In order to do this for a language with function symbols, we also need to define what it is for a given element d of a structure's domain to be the **denotation of** a term t **on** an assignment g.

- We write this as '$d = [\![t]\!]_S(g)$', or '$d = [\![t]\!]_S^g$'.

Definition of denotation on an assignment

Definition

Given a structure S for Σ, $[\![\cdot]\!]_S$ is the function from $\text{Terms}(\Sigma)$ to $D^{D^{\text{Var}}}$ such that

1. $[\![v]\!]_S^g = gv$ for any variable v.
2. $[\![c]\!]_S^g = I_c$ if c is an individual constant of Σ (for any g).
3. $[\![f(t_1, \ldots, t_n)]\!]_S^g = I_f([\![t_1]\!]_S^g, \ldots, [\![t_n]\!]_S^g)$

Definition of truth on an assignment

Given a structure S for Σ, the relation $S, g \Vdash P$ is the unique relation between assignment functions for S and formulae of Σ such that:

- $S, g \Vdash F(t_1, \ldots, t_n)$ iff $\langle [\![t_1]\!]_S^g, \ldots, [\![t_n]\!]_S^g \rangle \in I_F$.
- $S, g \Vdash t_1 = t_2$ iff $[\![t_1]\!]_S^g = [\![t_2]\!]_S^g$.
- $S, g \Vdash \neg P$ iff $S, g \nVdash P$.
- $S, g \Vdash P \to Q$ iff $S, g \nVdash P$ or $S, g \Vdash Q$.
- $S, g \Vdash P \wedge Q$ iff $S, g \Vdash P$ and $S, g \Vdash Q$.
- $S, g \Vdash P \vee Q$ iff either $S, g \Vdash P$ or $S, g \Vdash Q$.
- $S, g \Vdash \forall v P$ iff $S, g[v \mapsto d] \Vdash P$ for all d in the domain of S.
- $S, g \Vdash \exists v P$ iff $S, g[v \mapsto d] \Vdash P$ for some d in the domain of S.

Defining logical notions

Definition

Formula P of $\mathcal{L}(\Sigma)$ is **valid** iff P is true in every Σ-structure on every assignment.

Definition

Sequent $\Gamma \triangleright P$ of $\mathcal{L}(\Sigma)$ is **valid** iff for every Σ-structure S and every assignment g for S, if every member of Γ is true in S on g, then P is true in S on g.

Another way of saying that the sequent $\Gamma \triangleright P$ is valid is to say that P is a **logical consequence** of Γ, or in symbols, $\Gamma \vDash P$.

Definition

Set Γ of formulae of $\mathcal{L}(\Sigma)$ is **satisfiable** (logically consistent) iff there is a Σ-structure S and an assignment g for S such that every member of Γ is true in S on g.

Fact

$\Gamma \vDash P$ iff $\Gamma \cup \{\neg P\}$ is not satisfiable; Γ is satisfiable iff $\Gamma \nvDash \bot$ $[:= \neg\forall x(x = x))]$

Irrelevance Lemma for terms

If $gv = hv$ for all $v \in FV(t)$, then $[\![t]\!]_S^g = [\![t]\!]_S^h$.

Irrelevance Lemma for formulae

If $gv = hv$ for all $v \in FV(P)$, then $h \in [\![P]\!]_S$ if $g \in [\![P]\!]_S$.

Substitution Lemma for terms

$[\![t[s/v]]\!]_S^g = [\![t]\!]_S^{g[v \mapsto [\![s]\!]_S^g]}$.

Substitution Lemma for formulae

$S, g \Vdash P[s/v]$ iff $S, g[v \mapsto [\![s]\!]_S^g] \Vdash P$.

Soundness and Completeness

Two theorems

Two reason for being interested in models comes from the following key theorem:

Soundness Theorem

Whenever $\Gamma \vdash P$, $\Gamma \vDash P$.

Completeness Theorem

Whenever $\Gamma \vDash P$, $\Gamma \vdash P$.

Say that Γ is *consistent* iff there is some P such that $\Gamma \nvdash P$. Then the above claims are equivalent to:

Soundness Theorem (alternative form)

Every satisfiable set of formulae is consistent.

Completeness Theorem (alternativbe form)

Every consistent set of formulae is satisfiable.

The usefulness of the Soundness Theorem

Up to now we have had great ways of showing that sequents are provable (e.g. constructing a proof of them), but no good ways of showing that they *aren't* provable. We can look at a bunch of attempts and say 'This isn't a proof, and this isn't, and this isn't. . . '; but this procedure never rules out that there's a proof we haven't considered yet.

Thanks to the Soundness Theorem, we have a way of showing that $\Gamma \nvdash P$: we construct a structure S and assignment g such that every member of Γ is true in S on g, but P isn't true in S on g.

More on the philosophical significance of the Soundness Theorem

Some discussions suggest the Soundness Theorem is supposed to actually provide a justification for reasoning in accordance with the classical rules baked into $\vdash$. But this is a problematic thought for at least two separate reasons.

1. Like every proof we've ever done, the proof of the Soundness Theorem requires using many of these very rules in the metalanguage. If one were really worried about those rules, one wouldn't accept the proof.

2. The imagined justification would require, e.g. going from 'Sentence P is valid' to actually asserting a certain sentence P—accepting it as *really* true. But since there isn't a set that contains everything, it is obscure how such a transition would be justified.

Proving the Soundness Theorem

We're trying to show that *every provable sequent is valid* so of course what's needed is an *induction on provable sequents*. I'll do some cases and leave others as homework.

Assumption: Trivially $P \vDash P$: this just means for all S, g, if P is true in S on g, P is true in S on g.

Weakening: Suppose $\Gamma \vDash P$ and Δ is a set of formulae. We need to show $\Gamma, \Delta \vdash P$: in other words, for every structure S and assignment g for S, if every member of $\Gamma \cup \Delta$ is true in S on g, P is true in S on g. But if every member of $\Gamma \cup \Delta$ is true in S on g, every member of Γ is; so by our induction hypothesis, P is.

$\vee$**Intro1:** Suppose $\Gamma \vDash P$, and suppose every member of Γ is true in S on g. Then by the induction hypothesis, P is true in S on g, so either P is true in S on g or Q is, so by the definition of $[\![P \vee Q]\!]$, $P \vee Q$ is true in S on g.

$\vee$**Elim:** Suppose $\Gamma \vDash P \vee Q$, $\Gamma, P \vDash R$, and $\Gamma, Q \vDash R$. Suppose every member of Γ is true in S on g. Then by the IH, $P \vee Q$ is true in S on g, so by the definition of $[\![P \vee Q]\!]$, either P is true in S on g or Q is. In the former case, R is true in S on g by the second part of the IH; in the latter case, R is also true in S by the third part of the IH. So, R is true in S on g.

$\forall$**Intro:** Suppose $\Gamma \vDash P$ and v is a variable that isn't free in any member of Γ. Suppose every member of Γ is true in a certain S on a certain g. Let d be an arbitrary member of the domain of S. Since v isn't free in Γ, $g[v \mapsto d]$ agrees with g on $FV(\Gamma)$, so by the Irrelevance Lemma, every member of Γ is true in S on $g[v \mapsto d]$, so by the IH, P is true in S on $g[v \mapsto d]$. Since this holds for every d, we can conclude (looking at the definition of $[\![\forall v P]\!]$) that $\forall v P$ is true in S on g.

The Completeness Theorem

The Completeness Theorem

The Completeness Theorem

If $\Gamma \vDash P$, then $\Gamma \vdash P$.

Three key notions for this proof

Definition

Γ is **negation-complete** := for each formula P, either $P \in \Gamma$ or $\neg P \in \Gamma$.

Definition

Γ is **closed** := for each formula P, if $\Gamma \vdash P$, then $P \in \Gamma$.

Fact

If Γ is consistent and negation-complete, Γ is closed.

Proof: suppose is negation-complete, $\Gamma \vdash P$, but $P \notin \gamma$. Then $\neg P \in \Gamma$, so $\Gamma \vdash \neg P$, so Γ is inconsistent.

Definition

Γ is **witness-complete** := for each formula P and variable v, either $\forall v \neg P \in \Gamma$ or there is a term t such that $P[t/v] \in \Gamma$.

Strategy

Step Zero: every **negation-complete**, **witness-complete**, consistent set of formulae in the **identity-free** language $\mathcal{L}_{\neg,\wedge,\vee,\to,\forall,\exists}(\Sigma)$ is satisfiable.

Step One: every negation-complete, witness-complete, consistent set of formulae in $\mathcal{L}(\Sigma)$ is satisfiable.

Step Two: every witness-complete, consistent $\Gamma \subseteq \mathcal{L}(\Sigma)$ is a subset of some negation-complete, witness-complete, consistent Γ^+, and is thus satisfiable by Step One.

Step Three: every consistent $\Gamma \subseteq \mathcal{L}(\Sigma)$ in which countably infinitely many variables don't occur free is a subset of some witness-complete, consistent Γ^+, and is thus satisfiable by Step Two.

Step Four: every consistent $\Gamma \subseteq \mathcal{L}(\Sigma)$ can be turned by a relettering of free variables into one in which countably many infinitely many variables don't occur free, and is thus staisfiable by Step Three.

Step Zero: the identity-free language

Suppose Γ is a consistent, negation-complete, and witness-complete set of identity-free formulae of a signature Σ. Consider, the following structure S and assignment g:

$$
\begin{aligned}
D &:= \text{Terms}(\Sigma) \\
I_c &:= c \text{ for each individual constant of } \Sigma. \\
I_f(t_1, \ldots, t_n) &:= f(t_1, \ldots, t_n) \text{ for each } n\text{-ary function symbol } f \text{ of } \Sigma\texttt{\textbackslash y} \\
I_F &:= \{\langle t_1, \ldots, t_n\rangle \mid F(t_1, \ldots, t_n) \in \Gamma\} \text{ for each } n\text{-palce predicate } F \text{ of } \Sigma. \\
g(v) &:= v \text{ for each variable } v.
\end{aligned}
$$

We will prove that for all (identity-free) formulae P, $S, g \Vdash P$ iff $P \in \Gamma$.

Proof for Step Zero

First we need to show that $[\![t]\!]_S^g = t$ for every term t. This is a trivial induction.

Next, we show by induction on the construction of formulae that $S, g \Vdash P$ iff $P \in \Gamma$, for all P.

(i) Atomic formulae: $S, g \Vdash F(t_1, \ldots, t_n)$ iff $\langle [\![t_1]\!]_S^g, \ldots, [\![t_n]\!]_S^g \rangle \in I_F$, iff $\langle t_1, \ldots, t_n \rangle \in I_F$, iff $F(t_1, \ldots, t_n) \in \Gamma$.

(ii) Negation. Suppose $S, g \Vdash P$ iff $P \in \Gamma$. Then $S, g \Vdash \neg P$ iff $P \notin \Gamma$. But since Γ is consistent and negation-complete, $P \notin \Gamma$ iff $\neg P \in \Gamma$.

(ii) Conjunction. Suppose $S, g \Vdash P$ iff $P \in \Gamma$ and $S, g \Vdash Q$ iff $Q \in \Gamma$. Then $S, g \Vdash P \wedge Q$ iff $P \in \Gamma$ and $Q \in \Gamma$. But if $P \in \Gamma$ and $Q \in \Gamma$ we must have $P \wedge Q \in \Gamma$ (by closure, using $\wedge$Intro), and if $P \wedge Q \in \Gamma$ we must have $P \in \Gamma$ and $Q \in \Gamma$ (by closure, using $\wedge$Elim).

(iii) Universal quantification. Suppose as the induction hypothesis that $S, g \Vdash P[t/v]$ iff $P[t/v] \in \Gamma$. Suppose that $S, g \Vdash \forall vP$. Then $S, g[v \mapsto d] \Vdash P$ for all d in the domain, so by the Substitution Lemma, $S, g \Vdash P[t/v]$ for all t, so by the induction hypothesis, $P[t/v] \in \Gamma$ for all t, which given negation-completeness means that there is no t for which $\neg P[t/v] \in \Gamma$. Since Γ is witness-complete, it follows that $\forall vP \in \Gamma$.

Conversely, suppose that $\forall vP \in \Gamma$. Then by closure and $\forall$Elim, $P[t/v] \in \Gamma$ for all terms t, so by the induction hypothesis, $S, g \Vdash P[t/v]$ for all terms t. But then by the Substitution Lemma, $S, g[v \mapsto t] \Vdash P$ for all terms t, so $S, g \Vdash \forall vP$ (since everything in the domain of S is a term).

I'll leave the steps for $\vee, \rightarrow$, and $\exists$ as exercises.

Step One: adding identity

Once we add identity to the language, the structure that worked in Step Zero no longer does the job. Every atomic sentence of the form $t_1 = t_2$ where t_1 and t_2 are distinct terms is false on S on g. But a consistent Γ can of course contain some such formulae! To solve this, let's make the following definition:

Definition

Where t is any Σ-term, let $[t]_\Gamma$ be the set $\{s \mid t = s \in \Gamma\}$

Our new structure S' will have as its domain $\{[t]_\Gamma \mid t \in \text{Terms}(\Sigma)\}$.

And our new assignment g' will map each variable v to $[v]_\Gamma$.

Thanks to the =Intro and =Elim rules, we can prove the following (for a negation-complete, consistent Γ)

(a) $s = t \in \Gamma$ iff $[s]_\Gamma = [t]_\Gamma$.

Proof: Left to right: suppose $s = t \in \Gamma$ and $s = s' \in \Gamma$; then $t = s' \in \Gamma$ by =Elim and closure. Right to left: suppose $[s]_\Gamma = [t]_\gamma$. By =Intro and closure, $t = t \in \Gamma$, so $t \in [t]_\Gamma$, so $t \in [s]_\Gamma$, so $s = t \in \Gamma$.

(b) If $s_1 \in [t_1]_\Gamma$, and $\ldots s_n \in [t_n]_\Gamma$, then $[f(s_1, \ldots, s_n)]_\Gamma = [f(t_1, \ldots, t_n)]_\Gamma$

Proof: if the hypothesis is true, each $t_i = s_i \in \Gamma$. By =Intro and closure, $f(t_1, \ldots, t_n) = f(t_1, \ldots, t_n) \in \Gamma$. So by n applications of =Elim and closure, $f(s_1, \ldots, s_n) = f(t_1, \ldots, t_n) \in \Gamma$. The conclusion follows by part (a).

(c) If $s_1 \in [t_1]_\Gamma$, and $\ldots s_n \in [t_n]_\Gamma$, and $F(t_1, \ldots, t_n) \in \Gamma$, then $F(s_1, \ldots, s_n) \in \Gamma$.

Proof: by closure and =Elim.

So, we can coherently stipulate that the interpretatin functions of our new structure S' work as follows:

$$\begin{aligned} I_c &\coloneqq [c]_\Gamma \text{ for each individual constant } c. \\ I_f([t_1]_\Gamma, \ldots, [t_n]_\Gamma) &\coloneqq [f(t_1, \ldots, t_n)]_\Gamma \\ I_F &\coloneqq \{\langle [t_1]_\Gamma, \ldots, [t_n]_\Gamma \rangle \mid F(t_1, \ldots, t_n) \in \Gamma\} \end{aligned}$$

Another straightforward induction then proves that for every t, $[\![t]\!]^{g'}_{S'} = [t]_\Gamma$.

We can then redo the step for atomic formulae in the Step Zero proof:

$$\begin{aligned} S', g' \Vdash F(t_1, \ldots, t_n) &\text{ iff } \langle [\![t_1]\!]^{g'}_{S'}, \ldots, [\![t_n]\!]^{g'}_{S'} \rangle \in I_F \\ &\text{ iff } \langle [t_1]_\Gamma, \ldots, [t_n]_\Gamma \rangle \in I_F \\ &\text{ iff } F(t_1, \ldots, t_n) \in \Gamma \end{aligned}$$

And we also have atomic identity formulae.

$S', g' \Vdash s = t$ iff $[\![s]\!]^{g'}_{S'} = [\![t]\!]^{g'}_{S'}$, iff $[s]_\Gamma = [t]_\Gamma$, iff $s = t \in \Gamma$.

The rest of the Step Zero proof goes through just as before.

Step Two: sets that are witness complete but not negation complete

Extensibility Lemma

Every consistent Γ has a consistent, negation-complete extension (i.e. superset).

Note that if Γ is witness-complete, so are all of its extensions; so given Step One, this implies that every witness-complete consistent set is satisfiable.

Proving the Extensibility Lemma

There are countably infinitely many formulae; enumerate them as P_0, P_1, P_2. Define a countably infinite sequence of sets $\Gamma_0, \Gamma_1, \Gamma_2, \ldots$ recursively as follows:

$$\Gamma_0 = \Gamma$$

$$\Gamma_{n+1} = \begin{cases} \Gamma_n \cup \{P_n\} & \text{if this is consistent} \\ \Gamma_n \cup \{\neg P_n\} & \text{otherwise} \end{cases}$$

Finally let Γ^+ be $\bigcup_n \Gamma_n$.

Γ^+ is negation-complete.

Each Γ_n is consistent (induction on n, using $\neg$Intro and Cut to get that if $\Gamma_n \vdash \neg P_n$ and $\Gamma_n, \neg P_n \vdash \bot$, then $\Gamma_n \vdash \bot$.).

By the compactness of provability, this implies that Γ^+ is consistent.

Step Three: sets that aren't witness-complete

Say that Γ is *abstemious* iff there is a countably infinite set $v_1, v_2, \ldots$ of variables that aren't free in any element of Γ.

There are only countably many pairs $\langle P, u \rangle$ of a formula P and variable u. Enumerate them as $\langle P_1, u_1 \rangle, \langle P_2, u_2 \rangle, \ldots$. We define another sequence of extensions of Γ, as follows:

$$\Gamma^0 := \Gamma$$

$$\Gamma^{n+1} := \begin{cases} \Gamma^n \cup \{P_n[v_n/u_n]\} & \text{if this is consistent} \\ \Gamma^n \cup \{\forall u_n \neg P_n\} & \text{otherwise} \end{cases}$$

Define $\Gamma' = \bigcup_n \Gamma^n$.

Γ' is obviously witness-complete.

To show that it's consistent, we show that each Γ^n is consistent. But this follows from $\forall$Intro and Cut.

Step Four

Now we have that every abstemious, consistent set of formulae is satisfiable. What about the case of a non-abstemious set?

This is a little fiddly to work through, but what we do is to pick some function f from variables to variables whose range excludes countably infinitely many variables, and turn it into a function f^* on [sets of] formulae in the obvious way. We check that if $\Gamma \vdash P$ then $f^*[\Gamma] \vdash f^*P$, and conclude that if Γ is satsifiable, $f^*[\Gamma]$ is abstemious and satisfiable. So there's an S, g such that $S, g \Vdash f^*[\Gamma]$. Finally, if we let $g^*(v) = g(fv)$, it is straightforward to show that $S, g^* \Vdash P$ iff $S, g \Vdash f^*P$; thus $S, g^* \Vdash \Gamma$.